LED House — Electrical Shop
Use Case Table

Use Case 1: Register Customer Account

Use case name:	Register Customer Account	
Scenario:	A new visitor registers an account on the LED House website to be able to place orders.	
Triggering event:	A visitor clicks the 'Register' button on the homepage.	
Brief description:	The system allows a new user to create an account by providing personal details. Upon successful registration, the user can log in and access customer features.	
Actors:	End Customer (Primary)	
Related use cases:	UC02 – Customer Login, UC06 – Place Order	
Stakeholders:	End Customer, Shop Owner	
Preconditions:	The user does not already have an account. The registration page is accessible.	
Postconditions:	A new customer account is created and stored in the database. The customer receives a confirmation notification.	
Flow of activities:	**Actor**	**System**
	1. Navigates to the registration page. 2. Fills in full name, email address, phone number, and password. 3. Submits the registration form.	1. Displays the registration form. 2. Validates input fields (checks for duplicate email, required fields, password strength). 3. Creates the account and sends a confirmation email to the customer.
Exception conditions:	Email already in use: system displays an error message. Invalid input format: system highlights the problematic fields.	

Use Case 2: Customer Login

Use case name:	Customer Login
Scenario:	A registered customer logs into the LED House system to access their account and place orders.
Triggering event:	The customer clicks the 'Login' button on the homepage.
Brief description:	The system authenticates the customer using their registered email and password. On success, the customer is redirected to their account dashboard.

Actors:	End Customer (Primary)	
Related use cases:	UC01 – Register Customer Account, UC06 – Place Order	
Stakeholders:	End Customer, Shop Owner	
Preconditions:	The customer has a registered account. The login page is accessible.	
Postconditions:	The customer is authenticated and granted access to account features.	
Flow of activities:	**Actor**	**System**
	1. Navigates to the login page. 2. Enters email and password. 3. Clicks the 'Login' button.	1. Displays the login form. 2. Validates credentials against the database. 3. Grants access and redirects the customer to the dashboard.
Exception conditions:	Wrong credentials: system shows an error message. Account not found: system prompts registration.	

Use Case 3: Browse and Search Products

Use case name:	Browse and Search Products	
Scenario:	A customer browses the product catalogue or searches for a specific electrical product.	
Triggering event:	The customer lands on the product catalogue page or enters a keyword in the search bar.	
Brief description:	The system displays available products with filtering and search capabilities to help the customer find the desired item quickly.	
Actors:	End Customer (Primary)	
Related use cases:	UC04 – View Product Details, UC05 – Add to Cart	
Stakeholders:	End Customer, Shop Owner	
Preconditions:	Products are available in the system. The customer has internet access.	
Postconditions:	The customer views a filtered list of matching products.	
Flow of activities:	**Actor**	**System**
	1. Navigates to the catalogue or enters a search term. 2. Applies filters (category, price range, availability). 3. Reviews the search results.	1. Displays the full product catalogue. 2. Filters products based on search terms and applied filters. 3. Returns and displays the matching products with names, prices, and availability.

Exception conditions:	No matching products found: system displays a 'No results' message. Network error: system shows an error notification.

Use Case 4: View Product Details

Use case name:	View Product Details	
Scenario:	A customer clicks on a product to view its full details before deciding to purchase.	
Triggering event:	The customer clicks on a product listing from the catalogue or search results.	
Brief description:	The system presents the full product information including name, description, price, stock availability, and images.	
Actors:	End Customer (Primary)	
Related use cases:	UC03 – Browse and Search Products, UC05 – Add to Cart	
Stakeholders:	End Customer, Shop Owner	
Preconditions:	The product exists and is listed in the catalogue.	
Postconditions:	The customer views all details of the selected product.	
Flow of activities:	**Actor**	**System**
	1. Clicks on a product from the search or catalogue results. 2. Reviews product name, description, price, and availability.	1. Retrieves and displays the full product details from the database. 2. Shows stock status (in stock / out of stock).
Exception conditions:	Product no longer available: system shows 'Out of Stock' or redirects. Page load failure: system shows an error.	

Use Case 5: Add Product to Cart

Use case name:	Add Product to Cart
Scenario:	A logged-in customer adds a selected product to the shopping cart.
Triggering event:	The customer clicks the 'Add to Cart' button on a product details page.
Brief description:	The system adds the chosen product and quantity to the customer's shopping cart for later checkout.
Actors:	End Customer (Primary)
Related use cases:	UC04 – View Product Details, UC06 – Place Order
Stakeholders:	End Customer
Preconditions:	The customer is logged in. The product is in stock.

Postconditions:	The product is added to the cart with the selected quantity. Cart total is updated.	
Flow of activities:	**Actor**	**System**
	1. Selects the desired quantity. 2. Clicks 'Add to Cart'.	1. Validates product availability and quantity. 2. Adds the product to the customer's cart and updates the cart total.
Exception conditions:	Requested quantity exceeds stock: system alerts the customer. Customer not logged in: system redirects to login page.	

Use Case 6: Place Order

Use case name:	Place Order	
Scenario:	A customer proceeds from the shopping cart to place a final order for the selected products.	
Triggering event:	The customer clicks 'Proceed to Checkout' from the shopping cart.	
Brief description:	The system collects delivery information, confirms the cart contents, and processes the order. The order is saved with a 'Pending' status.	
Actors:	End Customer (Primary)	
Related use cases:	UC05 – Add to Cart, UC07 – Process Payment, UC10 – Receive Notification	
Stakeholders:	End Customer, Shop Owner	
Preconditions:	The customer is logged in. The shopping cart contains at least one product.	
Postconditions:	The order is saved in the database with 'Pending' status. Stock is reserved. Customer receives order confirmation.	
Flow of activities:	**Actor**	**System**
	1. Reviews cart contents. 2. Enters or confirms delivery address. 3. Selects payment method. 4. Confirms and places the order.	1. Displays cart summary and delivery form. 2. Validates delivery details. 3. Creates the order record with 'Pending' status. 4. Triggers payment processing and sends confirmation notification.
Exception conditions:	Payment failure: order is not confirmed, customer is notified. Stock unavailable at checkout: system alerts the customer.	

Use Case 7: Process Payment

Use case name:	Process Payment	
Scenario:	The system processes the customer's payment using the selected payment method.	
Triggering event:	The customer confirms the order and selects a payment method during checkout.	
Brief description:	The system handles payment through PayPal, 2Checkout (credit/debit card), or Cash on Delivery. For online payments, the customer is redirected to the external provider.	
Actors:	End Customer (Primary), Payment Provider (Secondary)	
Related use cases:	UC06 – Place Order, UC10 – Receive Notification	
Stakeholders:	End Customer, Bank / Payment Provider, Shop Owner	
Preconditions:	The order has been created. The customer has selected a payment method.	
Postconditions:	Payment is confirmed or declined. Order status is updated accordingly.	
Flow of activities:	**Actor**	**System**
	1. Selects PayPal, Credit/Debit Card, or Cash on Delivery. 2. Completes payment on the external provider page (if applicable). 3. Returns to LED House after payment.	1. Redirects to PayPal or 2Checkout as needed. 2. Receives payment confirmation from the provider. 3. Updates order status to 'Confirmed' or flags as failed.
Exception conditions:	Payment declined: system notifies the customer and keeps the order in 'Pending'. External provider unavailable: system suggests an alternative payment method.	

Use Case 8: Manage Products (Add/Update/Delete)

Use case name:	Manage Products (Add/Update/Delete)
Scenario:	The shop owner adds new products, updates existing ones, or removes discontinued items from the catalogue.
Triggering event:	The shop owner navigates to the product management section of the admin dashboard.
Brief description:	The system allows the shop owner to perform full CRUD operations on products, including setting name, description, price, stock quantity, and images.
Actors:	Shop Owner / Admin (Primary)
Related use cases:	UC09 – Manage Inventory, UC14 – View Sales Analytics

Stakeholders:	Shop Owner, Store Employee, End Customer	
Preconditions:	The shop owner is logged in with admin privileges.	
Postconditions:	Product catalogue is updated and changes are immediately reflected for customers.	
Flow of activities:	**Actor**	**System**
	1. Navigates to the product management dashboard. 2. Selects 'Add', 'Edit', or 'Delete' for a product. 3. Fills in or modifies product details and saves.	1. Displays the product list with management options. 2. Validates the entered data. 3. Saves changes to the database and updates the catalogue.
Exception conditions:	Missing required fields: system highlights errors. Delete a product with active orders: system prevents deletion and shows a warning.	

Use Case 9: Manage Inventory

Use case name:	Manage Inventory	
Scenario:	The system tracks product stock levels and updates them automatically after each purchase.	
Triggering event:	A customer successfully places an order, or the shop owner manually updates stock quantities.	
Brief description:	The system decrements stock after confirmed purchases and prevents orders for out-of-stock items. The owner can also manually adjust stock levels.	
Actors:	System (Primary), Shop Owner (Secondary)	
Related use cases:	UC06 – Place Order, UC08 – Manage Products	
Stakeholders:	Shop Owner, Store Employee, End Customer	
Preconditions:	Products are registered in the system with initial stock levels.	
Postconditions:	Stock is updated accurately. Out-of-stock products are flagged and prevented from being ordered.	
Flow of activities:	**Actor**	**System**
	1. Owner navigates to inventory management. 2. Adjusts stock quantities if needed.	1. Automatically deducts stock when an order is confirmed. 2. Flags products with zero stock as 'Out of Stock'. 3. Saves updated quantities to the database.

Exception conditions:	Stock goes negative due to concurrent orders: system applies a locking mechanism. Manual input error: system validates minimum stock value.

Use Case 10: Receive Notification

Use case name:	Receive Notification	
Scenario:	The customer is notified about updates to their order or service request status.	
Triggering event:	The order or service status changes (e.g., confirmed, delivered, repair ready).	
Brief description:	The system sends automated notifications to the customer via email or SMS when key status changes occur, keeping the customer informed throughout the process.	
Actors:	System (Primary), End Customer (Recipient)	
Related use cases:	UC06 – Place Order, UC11 – Submit Service Request, UC07 – Process Payment	
Stakeholders:	End Customer, Shop Owner	
Preconditions:	The customer has a registered account with a valid email or phone number. A status change event has occurred.	
Postconditions:	The customer has been notified. Notification log is stored in the system.	
Flow of activities:	**Actor**	**System**
	1. Customer receives a notification (email/SMS). 2. Customer reads the status update.	1. Detects the status change event. 2. Generates and sends the notification via email or SMS. 3. Logs the notification in the system.
Exception conditions:	Invalid email/phone: notification fails. System logs the failure and the owner is alerted to contact the customer manually.	

Use Case 11: Submit Service / Repair Request

Use case name:	Submit Service / Repair Request
Scenario:	A customer submits a repair or service request for an electrical product.
Triggering event:	The customer clicks 'Submit Service Request' from their account dashboard.
Brief description:	The customer describes the product and the issue. The system stores the request with a 'Pending' status and notifies the owner.

Actors:	End Customer (Primary)	
Related use cases:	UC12 – Update Service Status, UC10 – Receive Notification	
Stakeholders:	End Customer, Shop Owner, Store Employee	
Preconditions:	The customer is logged in.	
Postconditions:	The service request is stored in the database with 'Pending' status. The shop owner is notified.	
Flow of activities:	**Actor**	**System**
	1. Navigates to the service request section. 2. Provides product description and description of the issue. 3. Submits the request.	1. Displays the service request form. 2. Validates and saves the request. 3. Notifies the shop owner and sets status to 'Pending'.
Exception conditions:	Incomplete form submission: system highlights missing fields. System unavailable: request is not saved, customer is informed.	

Use Case 12: Update Service Request Status

Use case name:	Update Service Request Status	
Scenario:	The shop owner or employee updates the status of a customer service request.	
Triggering event:	The owner/employee navigates to the service management dashboard and selects a request.	
Brief description:	The owner or employee can change the status of a repair request to 'In Progress' or 'Ready' and optionally set an estimated completion time.	
Actors:	Shop Owner / Admin (Primary), Store Employee (Secondary)	
Related use cases:	UC11 – Submit Service Request, UC10 – Receive Notification	
Stakeholders:	Shop Owner, Store Employee, End Customer	
Preconditions:	A service request exists with 'Pending' or 'In Progress' status. The user is logged in with appropriate privileges.	
Postconditions:	Service status is updated. Customer is notified of the new status.	
Flow of activities:	**Actor**	**System**
	1. Navigates to the service management section. 2. Selects a service request.	1. Displays active service requests. 2. Saves the updated status. 3. Sends a notification to the customer about the status update.

	3. Updates the status and optionally sets an estimated completion date. 4. Saves changes.	
Exception conditions:	No service requests found: system shows an empty state message. Unauthorized user attempts update: system denies access.	

Use Case 13: Manage Customer Orders

Use case name:	Manage Customer Orders	
Scenario:	The shop owner views and manages incoming customer orders, updating their status as needed.	
Triggering event:	The shop owner navigates to the order management dashboard.	
Brief description:	The system presents all orders with their details and current statuses. The owner can update orders from 'Pending' to 'Confirmed' or 'Delivered'.	
Actors:	Shop Owner / Admin (Primary), Store Employee (Secondary)	
Related use cases:	UC06 – Place Order, UC10 – Receive Notification, UC09 – Manage Inventory	
Stakeholders:	Shop Owner, Store Employee, End Customer	
Preconditions:	At least one order exists in the system. The user is logged in with admin or employee privileges.	
Postconditions:	Order status is updated in the database. Customer is notified of the status change.	
Flow of activities:	**Actor**	**System**
	1. Navigates to the order management section. 2. Selects an order to view details. 3. Updates the order status (Confirmed / Delivered).	1. Retrieves and displays all orders with details. 2. Saves the new status to the database. 3. Triggers a notification to the customer.
Exception conditions:	Order not found: system shows a search error. Unauthorized status update attempt: system blocks action.	

Use Case 14: View Sales Analytics

Use case name:	View Sales Analytics	
Scenario:	The shop owner accesses the sales dashboard to review product performance and revenue.	
Triggering event:	The shop owner clicks on 'Sales Analytics' in the admin dashboard.	
Brief description:	The system displays sales metrics including total units sold, revenue, best-selling products, profit per product, and trends over a selected time period.	
Actors:	Shop Owner / Admin (Primary)	
Related use cases:	UC08 – Manage Products, UC15 – Analyze Profit	
Stakeholders:	Shop Owner	
Preconditions:	The owner is logged in with admin privileges. Sales data exists in the system.	
Postconditions:	The owner views the sales report. No data is modified.	
Flow of activities:	**Actor**	**System**
	1. Navigates to the Sales Analytics section. 2. Selects a time period for the report. 3. Reviews metrics and product performance data.	1. Retrieves sales data from the database for the selected period. 2. Calculates totals, rankings, and trends. 3. Displays charts and tables with product performance metrics.
Exception conditions:	No sales data for selected period: system shows an empty state. Report generation error: system logs the error and notifies the owner.	

Use Case 15: Analyze Profit

Use case name:	Analyze Profit
Scenario:	The shop owner analyzes profit margins for products by reviewing purchase costs vs. selling prices.
Triggering event:	The shop owner selects a product in the sales dashboard and clicks 'Analyze Profit'.
Brief description:	The system calculates and displays profit per unit, total profit, and profit margin for the selected product or for all products over a specified period.
Actors:	Shop Owner / Admin (Primary)
Related use cases:	UC14 – View Sales Analytics, UC08 – Manage Products
Stakeholders:	Shop Owner

Preconditions:	Products have purchase cost and selling price defined. The owner is logged in with admin privileges.	
Postconditions:	The owner views detailed profit analysis. No data is modified.	
Flow of activities:	**Actor**	**System**
	1. Navigates to the profit analysis section. 2. Selects a product or time period. 3. Reviews profit data and margin percentages.	1. Retrieves purchase cost and selling price for selected products. 2. Calculates profit per unit, total profit, and profit margin. 3. Displays results in a clear summary table or chart.
Exception conditions:	Purchase cost not set for a product: system flags it and excludes from calculation. No sales in selected period: system shows zero profit with a notice.	